HW 3 — Escape from a Dielectric & Jackson 7.12

How to read this: black = write this down · **red = the answer** · *teal = explanation, no need to copy* · grey box = optional

Every step is written as **From:** (a law) **Also:** (another fact) **We get:** (the result) — with a tag (Lec 4, slide 7) naming the source slide.

Problem 1 — Escape from a dielectric

> A point source of light is embedded near the flat surface of a dielectric of index n. Treat the emitted light as plane waves (rays) going out **isotropically**. Find the **fraction** of rays that can refract out into the vacuum above.

ANSWER

$$f = \frac{1}{2}\left[1 - \sqrt{1 - 1/n^2}\right]$$

Only rays inside a cone of half-angle i_o about the normal get out, where $\sin i_o = 1/n$.

*Glass ($n = 1.5$) gives **12.7%**; for large n it goes as **$1/4n^2$**.*

Step 0 — What do we actually need to know?

Two facts, and that is the whole problem. Here is where each one comes from.

From Lecture 4: Snell's law $n \sin i = n' \sin r$, with total internal reflection above $i_o = \arcsin(n'/n)$ (Lec 4, slide 7)

Also: the outside medium here is vacuum, so $n' = 1$

We get: $\sin i_o = 1/n$ *the critical angle for this problem*

From geometry: the solid-angle element is $d\Omega = \sin\theta\, d\theta\, d\phi$, and a whole sphere is 4π

Also: "isotropic" means **equal power into equal solid angle**

We get: fraction of rays = $\dfrac{\text{solid angle that escapes}}{4\pi}$

STARTING POINT — carry only these two lines into Step 1:

- a ray escapes only if it meets the surface at $\theta < i_o$, with $\sin i_o = 1/n$;
- $f = \Omega_{escape}/4\pi$.

Step 1 — Which rays can get out?

Label every ray by its angle θ from the normal. That angle is its angle of incidence when it reaches the surface.

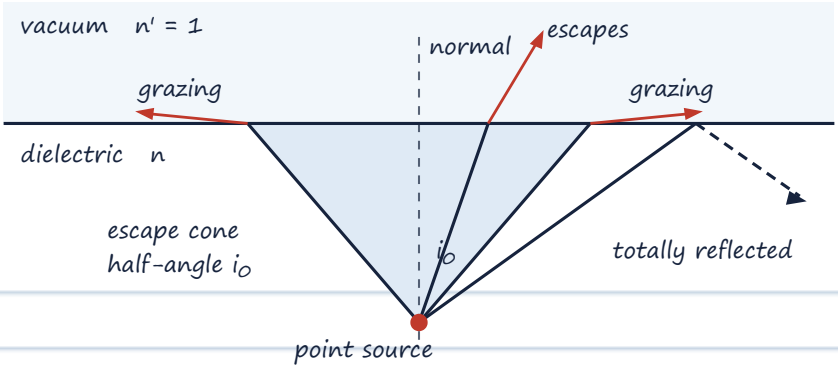
From Snell: $n \sin\theta = 1 \cdot \sin r \Rightarrow \sin r = n \sin\theta$

Also: $\sin r$ is the sine of a real angle, so it can never exceed 1

We get: a refracted ray exists only if $n \sin\theta \le 1$, i.e. $\theta < i_o \Rightarrow$ it escapes

We get: if $\theta > i_o$: $\sin r > 1$, $\cos r$ imaginary, evanescent only $\Rightarrow$ totally reflected

Lec 4, sl. 7–8



The escaping rays fill a cone of half-angle i_o around the upward normal.

Rays aimed downward never reach the surface at all, so they are out automatically. Only the upward cone counts.

Step 2 — How big is that cone?

Add up the solid angle of every direction with $\theta < i_o$

From Step 0: $d\Omega = \sin\theta\, d\theta\, d\phi$

Also: the escaping directions run $\theta = 0$ to i_o, and $\phi = 0$ all the way round to 2π

We get: $\Omega_{escape} = \int_0^{2\pi} d\phi \int_0^{i_o} \sin\theta\, d\theta = 2\pi\,[-\cos\theta]_0^{i_o} = 2\pi(1 - \cos i_o)$

Step 3 — Turn the cone into a fraction

From Step 0: $f = \Omega_{escape}/4\pi$

Also, from Step 2: $\Omega_{escape} = 2\pi(1 - \cos i_o)$

We get: $f = \dfrac{2\pi(1 - \cos i_o)}{4\pi} = \dfrac{1 - \cos i_o}{2}$

Step 4 — Write it with n instead of i_o

From Step 0: $\sin i_o = 1/n$

Also: $\sin^2 + \cos^2 = 1$

We get: $\cos i_o = \sqrt{1 - 1/n^2}$

FINAL ANSWER

$$f = \frac{1}{2}\left[\,1 - \sqrt{1 - 1/n^2}\,\right]$$

■ END OF REQUIRED SOLUTION — everything below is optional ■

OPTIONAL **Checks and numbers.**

- $n \to 1$ gives $i_o \to 90°$, $f \to ½$ — no index step, so everything heading upward escapes. ✓
- large n: $\sqrt{(1-1/n^2)} \approx 1 - 1/2n^2$, so $f \approx 1/4n^2$. ✓
- water 1.33 → 17.0% | glass 1.5 → 12.7% | diamond 2.4 → 4.5% | GaAs 3.5 → 2.1%
- Why LEDs need domes or roughened surfaces — a bare GaAs chip leaks only ~2%.
- We counted rays geometrically; inside the cone a ray is still partly reflected (Fresnel), so this f is an upper bound.

Problem 2 — Jackson 7.12

The time dependence of electrical disturbances in good conductors is governed by the frequency-dependent conductivity (7.58). Consider **longitudinal** electric fields in a conductor, using Ohm's law, the continuity equation, and the differential form of Coulomb's law.
(a) Show that the time-Fourier-transformed charge density satisfies $[\sigma(\omega) - i\omega\varepsilon_0]\rho(\mathbf{x},\omega) = 0$.
(b) Using $\sigma(\omega) = \sigma_0/(1 - i\omega\tau)$ with $\sigma_0 = \varepsilon_0\omega_p^2\tau$, show that for $\omega_p\tau \gg 1$ any initial disturbance oscillates at the plasma frequency and decays with constant $\lambda = 1/2\tau$.

ANSWER

(a) $[\sigma(\omega) - i\omega\varepsilon_0]\,\rho(\mathbf{x},\omega) = 0$

(b) the bracket vanishing gives $\omega^2 + i\omega/\tau - \omega_p^2 = 0$, so

$$\omega \approx \pm\omega_p - \frac{i}{2\tau} \quad \Rightarrow \quad \rho \propto e^{\mp i\omega_p t}\, e^{-t/2\tau}$$

— oscillates at ω_p, decays with $\lambda = 1/2\tau$.

Step 0 — The four things we are allowed to use

Three are EM I laws, one is from this semester. He has not worked this derivation in class, so write the four down and start from them.

From Ohm's law: $\vec{J} = \sigma(\omega)\vec{E}$ — EM I / Jackson (7.58)

Also, charge conservation: $\nabla\cdot\vec{J} + \partial\rho/\partial t = 0$ — the continuity equation

Also, Gauss's law: $\nabla\cdot\vec{E} = \rho/\varepsilon_0$ — Coulomb in differential form

Also, from Lecture 6: $\omega_p^2 = \dfrac{Ne^2}{\varepsilon_0 m}$ (Lec 6, high-frequency behavior)

One notation warning — his γ_0 is Jackson's $1/\tau$

On the Lecture 6 Metal slide he writes $\sigma_0 = Ne^2/(m\gamma_0)$; Jackson writes $\sigma_0 = \varepsilon_0\omega_p^2\tau = Ne^2\tau/m$. Same quantity $\Rightarrow$ $\tau = 1/\gamma_0$

Step 1 — Go to frequency space

Write every quantity as $f(t) = f(\omega)e^{-i\omega t}$ — the same convention used all semester. Then a time derivative becomes a multiplication.

From that convention: $\partial/\partial t \rightarrow -i\omega$

Also, continuity: $\nabla\cdot\vec{J} + \partial\rho/\partial t = 0$

We get: $\nabla\cdot\vec{J}(\mathbf{x},\omega) - i\omega\rho(\mathbf{x},\omega) = 0$

Step 2 — Substitute Ohm, then Gauss → part (a)

Two substitutions, one at a time. After both, only ρ is left in the equation.

From Step 1: $\nabla\cdot\vec{J} - i\omega\rho = 0$

Also, Ohm: $\vec{J} = \sigma(\omega)\vec{E}$

We get: $\sigma(\omega)\,\nabla\cdot\vec{E} - i\omega\rho = 0$

Also, Gauss: $\nabla\cdot\vec{E} = \rho/\varepsilon_0$

We get: $\dfrac{\sigma(\omega)}{\varepsilon_0}\,\rho - i\omega\rho = 0$

We get: multiply by ε_0, factor out ρ:

FINAL ANSWER — PART (a)

$$[\ \sigma(\omega) - i\omega\varepsilon_0\ ]\,\rho(\mathbf{x},\omega) = 0$$

Same thing in his notation. *Divide by $(-i\omega)$: $[\varepsilon_0 + i\sigma(\omega)/\omega]\rho = 0$, i.e. $\boldsymbol{\varepsilon_{eff}(\omega)\,\rho = 0}$ — the conductor permittivity $\varepsilon(\omega) = \varepsilon_r + ib/\omega$ from* Lec 7, slide 2 *set to zero. A charge wave lives only where the permittivity vanishes.*

Step 3 — Put in the given $\sigma(\omega)$

A real disturbance has $\rho \neq 0$, so the bracket itself must be zero. That becomes an equation for ω.

From part (a): $\rho \neq 0 \Rightarrow \sigma(\omega) - i\omega\varepsilon_0 = 0$

Also, given: $\sigma(\omega) = \dfrac{\sigma_0}{1 - i\omega\tau}$

We get: $\sigma_0 = i\omega\varepsilon_0(1 - i\omega\tau) = i\omega\varepsilon_0 + \omega^2\varepsilon_0\tau$

Also, given: $\sigma_0 = \varepsilon_0\omega_p{}^2\tau$

We get: $\varepsilon_0\omega_p{}^2\tau = i\omega\varepsilon_0 + \omega^2\varepsilon_0\tau$, then divide by $\varepsilon_0\tau$:

$$\omega^2 + \frac{i\omega}{\tau} - \omega_p{}^2 = 0$$

Step 4 — Solve that quadratic for ω

From the quadratic formula: with $a = 1$, $b = i/\tau$, $c = -\omega_p{}^2$

Also: $(i/\tau)^2 = -1/\tau^2$

We get: $\omega = -\dfrac{i}{2\tau} \pm \sqrt{\omega_p{}^2 - 1/4\tau^2}$

Step 5 — Use the given condition $\omega_p\tau \gg 1$

From $\omega_p\tau \gg 1$: $1/4\tau^2$ is negligible beside $\omega_p{}^2$

We get: $\omega \approx \pm\omega_p - \dfrac{i}{2\tau}$

Also: $\rho(t) \propto e^{-i\omega t}$

We get: $\rho(t) \propto e^{\mp i\omega_p t} \cdot e^{-t/2\tau}$ *real part → oscillation, imaginary part → decay*

FINAL ANSWER — PART (b)

oscillation at ω_p, decay constant $\lambda = \dfrac{1}{2\tau}$

■ END OF REQUIRED SOLUTION — everything below is optional ■

EXTRA — none of this page is required for the answer

Jackson's footnote: what goes wrong if you use σ_0

From part (a) with $\sigma = \sigma_0$: $[\sigma_0 - i\omega\varepsilon_0]\rho = 0$

We get: $\omega = -i\,\sigma_0/\varepsilon_0 \;\Rightarrow\; \rho \propto e^{-(\sigma_0/\varepsilon_0)t}$

*Purely imaginary $\omega \Rightarrow$ **no oscillation at all**, and the decay constant is $\lambda_w = \sigma_0/\varepsilon_0 = \omega_p^2\tau$. For $\omega_p\tau = 100$ that is 20000× faster than the true $1/2\tau$ — Jackson's "extremely rapid damping with the (wrong) decay constant". The lesson: the disturbance happens at $\omega \approx \omega_p$, where $\omega\tau \gg 1$, so the DC value of σ is the wrong one.*

OPTIONAL **The decay constant is exact.** The imaginary part of both roots is exactly $-1/2\tau$ whenever $\omega_p\tau > ½$ — only the oscillation frequency is approximated. For $\omega_p\tau = 100$ the exact roots are $\pm 99.9987/\tau - 0.5i/\tau$ against $\pm 100/\tau - 0.5i/\tau$.

OPTIONAL **Where ω_p came from.** Lecture 6 derives $\omega_p^2 = NZe^2/\varepsilon_0 m$ from the bound-electron model at high frequency, and quotes 60 MHz for the ionosphere ($N \sim 10^{12}/m^3$) and 6×10^{10}–6×10^{12} Hz for lab plasmas.